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A manufacturer produces a certain type of battery. The quality standards require the capacity of these batteries to be in between (137, 143) ampere-hours.

To meet these standards, the production process managers decided to design the production process such that:

1. Let μ be the population average of the battery capacities. The production process should ensure that $\mu = 140$.
2. Let σ^2 be the population variance of the battery capacities. The production process should ensure that $\sigma^2 = 4$.
3. Let p be the proportion of batteries that meet the quality standards. The production process should ensure that $p = 0.95$.

An independent consumer-protection agency wishes to test whether the manufacturer complies to the quality standards. For this purpose, they measure the capacity of a random sample of 20 batteries from a recently produced batch. They compiled the following information from this sample.

Note that the random sample consists of $(X_1, X_2, \dots, X_n)$:

- X_i denotes the capacity of i^{th} battery.
- Y denotes the number of batteries complying the quality standards.
- n is the sample size.

Assume that all the underlying assumptions for the confidence intervals given above are verified by the agency. Hence, $(X_1, X_2, \dots, X_n)$ are normal with the same parameters and independent.

Parameter	μ	σ^2	p
Point estimator	Sample capacity mean, $\bar{X}$	Sample capacity variance, S^2	$\hat{p} = \frac{Y}{n}$
Point estimate	$\bar{x} = 138.47$	$s^2 = (2.66)^2 = 7.08$	$\hat{p} = \frac{y}{n} = 0.65$
Standard error of the point estimator	$\sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}}$	No closed-form expression	$\sigma_{\hat{p}} = \sqrt{\frac{p(1-p)}{n}}$
Estimated standard error of the point estimator	$\hat{\sigma}_{\bar{X}} = \frac{s}{\sqrt{n}} = 0.595$	No straightforward computation	$\hat{\sigma}_{\hat{p}} = \sqrt{\frac{\hat{p}(1-\hat{p})}{n}} = 0.11$
Expression for 95% CI	$(\bar{x} \mp t_{0.025,19} \hat{\sigma}_{\bar{X}})$	$\left(\frac{(n-1)s^2}{\chi_{n-1,\alpha/2}^2}, \frac{(n-1)s^2}{\chi_{n-1,1-\alpha/2}^2} \right)$	$(\hat{p} \mp z_{0.025} \hat{\sigma}_{\hat{p}})$
95% CI	(137.23, 139.71)	(4.09, 15.08)	(0.44, 0.86)
95% PI	(135.16, 141.78)	--	--
95% TI for 90%	(123.32, 153.62)	--	--

Now use the information in the above table to answer the following questions:

1. Does the manufacturer meet the quality standards? Why or why not? Explain.
2. Does the production process satisfy the managers' design? Why or why not? Explain.
3. If the production process satisfied the managers' design, do you think it would also meet the quality standards? Why or why not? Explain.

1. The manufacturer does not meet the quality standards:
 - a. The 95% CI around the mean (137.23, 139.71) is in the interval for the quality standards (137, 143). However, this is only for the population mean. Hence, it does not have an implication about meeting the quality standards alone. To understand the extent that the batteries meet the standards, we need to check PI and TI.
 - b. The 95% PI is (135.16, 141.78) so that quite a few of batteries may have a capacity below 137. This indicates that batteries do not meet the standards at a high level.
 - c. The 95% TI for 90% coverage is (123.32, 153.62). Hence, we expect to have 90% of the population within this interval with 95% confidence. This interval is much larger than (137, 143). Ideally, we would have liked this interval to lay within (137, 143). We could have been satisfied, if the two intervals were closer to each other, but in this case, they are very different. This also shows that the quality standards are not met.
2. The production process does not satisfy the design:
 - a. The 95% CI around the mean (137.23, 139.71) does not include the target mean level 140.
 - b. The 95% CI around the variance of the capacity is (4.09, 15.08) does not include the target variance 4.
 - c. The 95% CI around the proportion of batteries that satisfy the quality standards is (0.44, 0.86) does not include the target proportion 95%.
3. If the production process satisfied the managers' design, it would not meet the quality standards:
 - a. Let us take the two targets $\mu = 140$ and $\sigma^2 = 4$. Now we calculate the corresponding p, proportion of batteries that satisfy the quality standards.

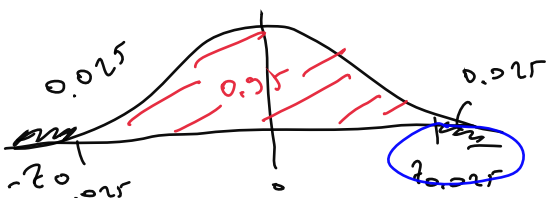
In this case, $X_i \sim N(140, 4)$. Then:

$$\begin{aligned}
 P(\text{battery satisfies q.s.}) &= P(137 < X_i < 143) = P\left(\frac{137-140}{2} < \frac{X_i-140}{2} < \frac{143-140}{2}\right) \\
 &= P(-1.5 < Z < 1.5) = \Phi(1.5) - \Phi(-1.5) = 2\Phi(1.5) - 1 = 2 \times 0.9332 - 1 \\
 &= 86.64\% < 95\%
 \end{aligned}$$

- b. Let us take the two targets $\mu = 140$ and $p=95\%$. Now we calculate the corresponding σ^2 , variance of the battery capacities.

In this case, $X_i \sim N(\mu=140, \sigma^2)$. We would like to have:

$$\begin{aligned}
 0.95 &= P(\text{battery satisfies q.s.}) = P(137 < X_i < 143) \\
 &= P\left(\frac{137-140}{\sigma} < \frac{X_i-140}{\sigma} \leq \frac{143-140}{\sigma}\right) \\
 \Rightarrow z_{0.025} &= 1.96 = \frac{143-140}{\sigma}
 \end{aligned}$$



$$\Rightarrow \sigma = \frac{3}{1.96} = 1.53 < 2.$$